13.7

Date:

solve

Let $f = x^2 + y^2 + z^2 - 25 = 0$

$f_x = 2x$, $f_y = 2y$, $f_z = 2z$ at $P(-3, 0, 4)$

$a = -6$, $b = 0$, $c = 8$

- Tangent Plane:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$
$$-6(x+3) + 0(y-0) + 8(z-4) = 0$$
$$-6x - 18 + 8z - 32 = 0$$
$$-6x + 8z - 50 = 0$$
$$3x - 4z = -25$$

- Normal line:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

$$\frac{x+3}{-6} = \frac{y-0}{0} = \frac{z-4}{8}$$

4. $x^2y - 4z^2 = -7$; $P(-3, 1, -2)$

solve

Let $f = x^2y - 4z^2 + 7 = 0$

$f_x = 2xy$, $f_y = x^2$, $f_z = -8z$ at $P(-3, 1, -2)$

$a = -6$, $b = 9$, $c = 16$

- Tangent Plane:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

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$$-6(x+3)+9(y-1)+16(z+2)=0$$
$$-6x-18+9y-9+16z+32=0$$
$$-6x+9y+16z=-5.$$

- Normal line (Symetric Equations):

$$\frac{x-x_0}{a}=\frac{y-y_0}{b}=\frac{z-z_0}{c}$$

$$\frac{x+3}{-6}=\frac{y-1}{9}=\frac{z+2}{16}$$

5. $x^2-xyz=56$; $P(-4,5,2)$

Sol:

$$f(x,y,z)=x^2-xyz-56=0$$

$$f_x=2x-yz \quad , \quad f_y=-xz \quad , \quad f_z=-xy \quad \text{at } P(-4,5,2)$$

$$a=2(-4)-(5)(2)=-18$$
$$b=-(-4)(2)=8$$
$$c=-(-4)(5)=20$$

- Tangent Plane:

$$a(x-x_0)+b(y-y_0)+c(z-z_0)=0$$
$$-18(x+4)+8(y-5)+20(z-2)=0$$
$$-18x-72+8y-40+20z-40=0$$
$$-18x+8y+20z=152$$

- Normal Line:

$$\frac{x+4}{-18}=\frac{y-5}{8}=\frac{z-2}{20}$$

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6. $z = x^2 + y^2$; $P(2, -3, 13)$

solve

$f(x,y,z) = x^2 + y^2 - z = 0$

$f_x = 2x$, $f_y = 2y$, $f_z = -1$ at $P(2, -3, 13)$

$a = 2(2) = 4$, $b = 2(-3) = -6$, $c = -1$

- Tangent Plane:

$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$
$4(x - 2) + (-6)(y + 3) + (-1)(z - 13) = 0$
$4x - 8 - 6y - 18 - z + 13 = 0$
$4x - 6y - z = 13$

- Normal line:

$$\frac{x-2}{4} = \frac{y+3}{-6} = \frac{z-13}{-1}$$

7. $z = 4x^3y^2 + 2y$; $P(1, -2, 12)$

solve

$f(x,y,z) = 4x^3y^2 + 2y - z$

$f_x = 12x^2y^2$, $f_y = 8x^3y + 2$, $f_z = -1$ at $P(1, -2, 12)$

$a = 12(1)^2(-2)^2 = 48$
$b = 8(1)^3(-2) + 2 = -14$
$c = -1$

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• Tangent Plane:

$$a(x-x_0)+b(y-y_0)+c(z-z_0)=0$$
$$48(x-1)+(-14)(y+2)-1(z-12)=0$$
$$48x-48-14y-28-z+12=0$$
$$48x-14y-z=64$$

• Normal Line:

$$\frac{x-1}{48}=\frac{y+2}{-14}=\frac{z-12}{-1}$$

8 $z=\frac{1}{2}x^7y^{-2}$; $P(2,4,4)$

solve $f(x,y,z)=\frac{1}{2}x^7y^{-2}-z=0$

$f_x=\frac{7}{2}x^6y^{-2}$, $f_y=-x^7y^{-3}$, $f_z=-1$ at $P(2,4,4)$

$$a=7/2\,(2)^6(4)^{-2}=\frac{7(64)}{2(16)}=14$$
$$b=-(2)^7(4)^{-3}=\frac{-128}{64}=-2$$
$$c=-1$$

• Tangent Plane:

$$a(x-x_0)+b(y-y_0)+c(z-z_0)=0$$
$$14(x-2)-2(y-4)-1(z-4)=0$$
$$14x-28-2y+8-z+4=0$$
$$14x-2y-z=16$$

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- Normal line:

$$\frac{x-2}{14} = \frac{y-4}{-2} = \frac{z-4}{-1}$$

9. $z = xe^{-y}$; $P(1,0,1)$

$$f(x,y,z) = xe^{-y} - z = 0$$

$$f_x = e^{-y}, \quad f_y = -xe^{-y}, \quad f_z = -1 \text{ at } P(1,0,1)$$

$$a = e^0 = 1, \quad b = -(1)e^0 = -1, \quad c = -1$$

- Tangent line:

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$
$$1(x-1) - 1(y-0) - 1(z-1) = 0$$
$$x - 1 - y - z + 1 = 0$$
$$x - y - z = 0$$

- Normal line:

$$\frac{x-1}{1} = \frac{y-0}{-1} = \frac{z-1}{-1}$$

10. $z = \ln\sqrt{x^2+y^2}$; $P(-1,0,0)$

$$f(x,y,z) = \frac{1}{2}\ln(x^2+y^2) - z = 0$$

$$f_x = \frac{1}{2} \cdot \frac{2x}{x^2+y^2} = \frac{x}{x^2+y^2}$$

13.7

Date

$$f_y = \frac{1}{x} \cdot \frac{xy}{x^2+y^2} = \frac{y}{x^2+y^2}$$

$$f_z = -1$$

$$a = -1, \quad b = 0, \quad c = -1$$

- Tangent Plane:

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$
$$-1(x+1) + 0(y-0) - 1(z-0) = 0$$
$$-x - 1 - z = 0$$
$$x + z = -1$$

- Normal line:

$$\frac{x-1}{-1} = \frac{y-0}{0} = \frac{z-0}{-1}$$

11. $z = e^{3y} \sin(3x)$; $P(\pi/6, 0, 1)$

solution

$$f(x,y,z) = e^{3y} \sin(3x) - z = 0$$

$$f_x = 3e^{3y} \cos(3x), \quad f_y = 3e^{3y} \sin(3x), \quad f_z = -1$$

$$a = 3e^0 \cos(\pi/2) = 3(1)(0) = 0$$
$$b = 3e^0 \sin(\pi/2) = 3(1)(1) = 3$$
$$c = -1$$

- Tangent Plane:

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$a(x-x_0)+b(y-y_0)+c(z-z_0)=0$

$0(x-\pi/6)+3(y-0)-1(z-1)=0$

$3y-z+1=0$

$3y-z=-1$

- Normal Line.

$$\frac{x-\pi/6}{0}=\frac{y-0}{3}=\frac{z-1}{-1}$$

12. $z=x^{\frac{1}{2}}+y^{\frac{1}{2}}$, $P(4,9,5)$

solve

$f(x,y,z)=x^{\frac{1}{2}}+y^{\frac{1}{2}}-z=0$

$f_x=\frac{1}{2\sqrt{x}}$, $f_y=\frac{1}{2\sqrt{y}}$, $f_z=-1$

$a=1/4$ $b=1/6$, $c=-1$

- Tangent Plane:

$a(x-x_0)+b(y-y_0)+c(z-z_0)=0$

$\frac{1}{4}(x-4)+\frac{1}{6}(y-9)-1(z-5)=0$

Multiplying by 12 to clear fractions.

$3(x-4)+2(y-9)-12(z-5)=0$

$3x-12+2y-18-12z+60=0$

$3x+2y-12z=-30$

- Normal Line:

$$\frac{x-4}{3}=\frac{y-9}{2}=\frac{z-5}{-12}$$

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Ex #13.8

Date

Maxima/Minima

$f'(x) = 0$

$f_x = 0 \quad , \quad f_y = 0$

$f''(x)$

$A = f_{xx} \quad , \quad B = f_{xy} \quad ; \quad C = f_{yy}$

$f''(x) > 0 \rightarrow +ve$ minima

$f''(x) < 0 \rightarrow -ve$ maxima

$f''(x) = 0$ 1st Case Fail

$\boxed{D = AC - B^2}$

1) $D > 0 \quad , A > 0 \longrightarrow$ Minima
2) $D > 0 \quad , A < 0 \longrightarrow$ Maxima
3) $D < 0 \longrightarrow$ Saddle Point
4) $D = 0 \longrightarrow$ No Conclusion

1-2 Locate all absolute maxima and minima, if any by inspection. Then check your answers using calculus.

1.a) $f(x,y) = (x-2)^2 + (y+1)^2$

Solution

Partial Derivative:

$$f_x = 2(x-2) \quad , \quad f_y = 2(y+1)$$

Critical Points:

$f_x = 0$ | $f_y = 0$

$2(x-2) = 0$ | $2(y+1) = 0$

$x - 2 = 0$ | $y + 1 = 0$

$x = 2$ | $y = -1$

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Minimum at (2, -1), No maxima.

Sum of square is smallest at its critical point.

b) $f(x,y) = 1 - x^2 - y^2$

solve

$f_x = -2x$, $f_y = -2y$

$-2x = 0$ $\quad$ $-2y = 0$

$x = 0$ $\quad$ $y = 0$

Maxima at (0, 0), no minima.
Reason: Function is $1-(x^2+y^2)$, which is largest when $x, y = 0$.

c) $f(x,y) = x + 2y - 5$

solve

$f_x = 1$, $f_y = 2$

No maxima or minima.
Reason: No critical points.

2. a) $f(x,y) = 1 - (x+1)^2 - (y-5)^2$

solve

$f_x = -2(x+1)$, $f_y = -2(y-5)$

$f_x = 0$ $\quad$ $f_y = 0$

$-2(x+1) = 0$ $\quad$ $-2(y-5) = 0$

$x + 1 = 0$ $\quad$ $y - 5 = 0$

$x = -1$ $\quad$ $y = 5$

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Maxima at $(-1, 5)$, no minima.
Reason: Function is 1 - (sum of squares), which is largest when squares are zero.

b) $f(x,y) = e^{xy}$

solve

$f_x = ye^{xy}$, $f_y = xe^{xy}$

$f_x = 0$, $f_y = 0$
$ye^{xy} = 0$, $xe^{xy} = 0$
$y = 0$, $x = 0$

c) $f(x,y) = x^2 - y^2$

solve

$f_x = 2x$, $f_y = -2y$

$f_x = 0$, $f_y = 0$
$2x = 0$, $-2y = 0$
$x = 0$, $y = 0$

9-20 Locate all relative maxima, relative minima, and saddle points, if any.

9. $f(x,y) = y^2 + xy + 3y + 2x + 3$

solve

1) First Partial Derivatives:

$f_x = y + 2$

$f_y = 2y + x + 3$

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2) Find Critical Points:

Set $f_x = 0 \Rightarrow y + 2 = 0 \Rightarrow \boxed{y = -2}$

Substitute $y = -2$ in $f_y = 0$

$$2(-2) + x + 3 = 0$$
$$-4 + x + 3 = 0$$
$$x - 1 = 0$$
$$\boxed{x = 1}$$

The only critical point $(1, -2)$.

3) Second Partial Derivatives.

$$f_{xx} = 0 \quad , f_{yy} = 2 \quad , f_{xy} = 1$$

4) Calculate Discriminant (D):

$$D = (f_{xx})(f_{yy}) - (f_{xy})^2$$
$$D = (0)(2) - (1)^2 = -1$$

5) Conclusion :-

Since $D < 0$, the point $(1, -2)$ is a saddle point.

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10. $f(x,y) = x^2 + xy - 2y - 2x + 1$

Solve

First Partial Derivative

$f_x = 2x + y - 2$, $f_y = x - 2$

For Critical Points.

Set $f_y = 0 \Rightarrow x - 2 = 0 \Rightarrow \boxed{x = 2}$

Substitute $x = 2$ into $f_x = 0$

$$2(2) + y - 2 = 0$$
$$4 + y - 2 = 0$$
$$y = -2$$

The critical point is $(2, -2)$.

Second Partial Derivatives.

$f_{xx} = 2$, $f_{yy} = 0$, $f_{xy} = 1$

Calculate Discriminant (D):

$$D = (f_{xx})(f_{yy}) - (f_{xy})^2$$

$$D = (2)(0) - (1)^2 = -1$$

Conclusions:

Since $D < 0$, the point $(2, -2)$ is a saddle point.

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11. $f(x,y) = x^2 + xy + y^2 - 3x$

Solve

First Partial Derivatives.

$f_x = 2x + y - 3$, $f_y = x + 2y$

Find Critical Points.

From $f_y = 0$, we get $x = -2y$.

Substitute $x = -2y$ into $f_x = 0$

$2(-2y) + y - 3 = 0$

$-4y + y - 3 = 0$

$-3y = 3$

$\boxed{y = -1}$

$x = -2(-1)$

$\boxed{x = 2}$

The only critical point is $(2, -1)$.

Second Partial Derivatives:

$f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = 1$

$a = 2$ $\quad c = 2$ $\quad b = 1$

Calculate Discriminant (D):

$D = ac - b^2$

$D = (2)(2) - (1)^2 = 4 - 1 = 3$

Since $D = 3 > 0$ and $a = 2 > 0$, the point $(2, -1)$ is a relative minimum.

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12. $f(x,y) = xy - x^3 - y^2$

First Partial Derivatives:

$f_x = y - 3x^2$, $f_y = x - 2y$

Find Critical Points:

From $f_y = 0$, we get $x = 2y$.

Substitute $x = 2y$ into $f_x = 0$

$y - 3(2y)^2 = 0$
$y - 3(4y^2) = 0$
$y - 12y^2 = 0$
$y(1 - 12y) = 0$

$y = 0$	$1 - 12y = 0$
	$y = 1/12$
$x = 2y$	$x = 2\left(\frac{1}{12}\right)$
$x = 2(0)$	
$x = 0$	$x = \frac{1}{6}$

Second Partial Derivatives.

$f_{xx} = -6x$, $f_{yy} = -2$, $f_{xy} = 1$
$A = -6x$, $C = -2$, $B = 1$

$D = AC - B^2$

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$D = (-6x)(-2) - (1)^2 = 12x - 1$

For $(0,0)$: $D = 12(0) - 1 = -1$

$D < 0$, the point $(0,0)$ is a saddle point.

For $(1/6, 1/12)$: $D = 12(1/6) - 1 = 2 - 1 = 1$.

$D > 0$ and $A < 0$, the point $(1/6, 1/12)$ is a relative maximum.

13. $f(x,y) = x^2 + y^2 + \frac{2}{xy}$

Solve

First Partial Derivatives:

$f(x,y) = x^2 + y^2 + 2x^{-1}y^{-1}$

$f_x = 2x - 2x^{-2}y^{-1} = 2x - \frac{2}{x^2y}$

$f_y = 2y - 2x^{-1}y^{-2} = 2y - \frac{2}{xy^2}$

Find Critical Points..

$f_x = 0 \Rightarrow 2x = \frac{2}{x^2y} \Rightarrow x^3y = 1$

$f_y = 0 \Rightarrow 2y = \frac{2}{xy^2} \Rightarrow xy^3 = 1$

$x^3y = 1$

$\left(\frac{1}{y^3}\right)^3 y = 1$

$xy^3 = 1$

$x = \frac{1}{y^3}$

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$\frac{1}{y^8} = 1$

$y^8 = 1$

$y = \pm 1$

Put $y = 1$

$x = \frac{1}{(1)^3} = 1$

Put $y = -1$

$x = \frac{1}{(-1)^3} = -1$

Points = $(1, 1)$ & $(-1, -1)$.

Second Partial Derivative:

$f_{xx} = 2 + 4x^{-3}y^{-1} = 2 + \frac{4}{x^3y}$

$f_{xy} = 2x^{-2}y^{-2} = \frac{2}{x^2y^2}$

$f_{yy} = 2 + 4x^{-1}y^{-3} = 2 + \frac{4}{xy^3}$

Point $(1, 1)$	Point $(-1, -1)$
$A = 6$	$A = 6$
$B = -2$	$B = -2$
$C = 6$	$C = 6$
$D = AC - B^2$	$D = AC - B^2$
$D = (6)(6) - (-2)^2$	$D = (6)(6) - (-2)^2$
$D = 32 > 0, A > 0$	$D = 32 > 0, A > 0$
The point $(-1, -1)$ is a relative minimum.	The point $(-1, -1)$ is a relative minimum

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13.9

Date 7/11/2025

Lagrange Multiplier :-

Surface $f = f(x, y)$

Subject to $g = g(x, y)$

$\Rightarrow \nabla f = \lambda \nabla g$

$(f_x, f_y) = \lambda (g_x, g_y)$

$$\Rightarrow \begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \end{cases}$$

Example 1 At what point or points on the circle $x^2 + y^2 = 1$ does $f(x, y) = xy$ have an absolute maximum, and what is the maximum?

Solve

$\nabla f = \lambda \nabla g$

$\Rightarrow f_x = \lambda g_x$, $f_y = \lambda g_y$

$\Rightarrow y = \lambda 2x$, $x = \lambda 2y$

$\Rightarrow \lambda = \frac{y}{2x}$, $\lambda = \frac{x}{2y}$

$\Rightarrow \lambda = \lambda$

$\Rightarrow \frac{x}{2y} = \frac{y}{2x}$

$\Rightarrow x^2 = y^2 \quad \Rightarrow y = \pm x$

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Case A: $y = x$

Putting $y = x$ in constraint function $g(x,y) = x^2 + y^2 - 1$.

$$x^2 + x^2 = 1$$
$$2x^2 = 1$$
$$x^2 = 1/2$$
$$x = \pm \frac{1}{\sqrt{2}}$$

This gives 2 points: $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and $\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$

Case B: $y = -x$

Substitute $y = -x$ into $x^2 + y^2 = 1$

$$x^2 + (-x)^2 = 1$$
$$2x^2 = 1$$
$$x = \pm \frac{1}{\sqrt{2}}$$

This gives two more points: $\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$ and $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.

- Plug the points into the original objective function $f(x,y) = xy$, to see the maxima & minima.

(x,y)	$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$	$\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$	$\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$	$\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
$f(x,y) = xy$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$

maxima — minima

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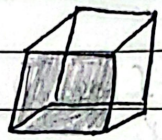
Q4 Use Lagrange multipliers to determine the dimensions of a rectangular box, open at the top, having a volume of $32\,ft^3$, and requiring the least amount of material for its construction.

Sol:



$$g = V = xyz = 32 \rightarrow \text{Constraint}$$

$$f = S = xy + 2yz + 2zx \rightarrow \text{Objective}$$

$$\nabla f = \lambda \nabla g$$

$$f_x = \lambda g_x \quad , \quad f_y = \lambda g_y \quad , \quad f_z = \lambda g_z$$

$$\Rightarrow y + 2z = \lambda yz \quad , \quad x + 2z = \lambda xz \quad , \quad 2y + 2x = \lambda xy$$

$$\Rightarrow \frac{1}{z} + \frac{2}{y} = \lambda \quad , \quad \frac{1}{z} + \frac{2}{x} = \lambda \quad , \quad \frac{2}{x} + \frac{2}{y} = \lambda$$

$$\Rightarrow \lambda = \lambda$$

$$\Rightarrow \frac{1}{z} + \frac{2}{y} = \frac{1}{z} + \frac{2}{x}$$

$$\Rightarrow x = y$$

Now set (2) = (3)

$$\frac{1}{z} + \frac{2}{x} = \frac{2}{x} + \frac{2}{y}$$

$$y = 2z$$

$$z = \frac{y}{2}$$

$$xyz = 32$$

$$(y)(y)\left(\frac{y}{2}\right) = 32$$

$$y^3 = 64$$

$$\sqrt[3]{y^3} = \sqrt[3]{64}$$

$$\boxed{y = 4}$$

$$\boxed{x = 4}$$

$$z = \frac{4}{2} \quad \boxed{= 2}$$

The least amount of material required when the dimensions of the box are 4ft by 4ft by 2Ft.

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5-12 Use Lagrange multipliers to find the maximum and minimum values of f subject to the given constraint. Also, find the points at which these extreme values occur.

5. $f(x,y) = xy \; ; \; 4x^2 + 8y^2 = 16$

solve

$f = xy \longrightarrow (1)$

$g = 4x^2 + 8y^2 - 16 \longrightarrow (2)$

$\nabla f = \lambda \nabla g$

$\Rightarrow f_x = \lambda g_x \quad , \quad f_y = \lambda g_y$

$\Rightarrow y = \lambda 8x \quad , \quad x = \lambda 16y$

$\Rightarrow \lambda = \frac{y}{8x} \quad , \quad \lambda = \frac{x}{16y}$

$\Rightarrow \lambda = \lambda$

$\Rightarrow \frac{y}{8x} = \frac{x}{16y}$

$\Rightarrow 2y^2 = x^2$

$\Rightarrow x = \pm y\sqrt{2}$

Substitute $x^2 = 2y^2$ in eqn (2).

$4(2y^2) + 8y^2 = 16$

$8y^2 + 8y^2 = 16$

$16y^2 = 16$

$y^2 = 1 \Rightarrow \boxed{y = \pm 1}$

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∴ Now find x :

$$x^2 = 2(1) = 2$$
$$x = \pm\sqrt{2}$$

(x, y)	$(1, \sqrt{2})$	$(1, -\sqrt{2})$	$(-1, \sqrt{2})$	$(-1, -\sqrt{2})$
$f(x,y) = xy$	$\sqrt{2}$	$-\sqrt{2}$	$-\sqrt{2}$	$\sqrt{2}$

Minimum

Maximum

6. $f(x, y) = x^2 - y^2$; $x^2 + y^2 = 25$

solve

$$g = x^2 + y^2 - 25$$

$$\nabla f = \lambda \nabla g$$
$$f_x = \lambda g_x \quad , \quad f_y = \lambda g_y$$

$$2x = \lambda 2x \quad , \quad -2y = \lambda 2y$$
$$\lambda = 1 \quad , \quad \lambda = -1$$

No constraint possible.

7. $f(x, y) = 4x^3 + y^2$; $2x^2 + y^2 = 1$

solve

$$g = 2x^2 + y^2 - 1$$

$$\nabla f = \lambda \nabla g$$

$$f_x = \lambda g_x \quad , \quad f_y = \lambda g_y$$

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$12x^2 = \lambda 4x$, $2y = \lambda 2y$

$3x = \lambda$, $\lambda = 1$

$2y = \lambda 2y$

$\lambda = \lambda$ $\quad 2y(1-\lambda) = 0$

$3x = 1$, $y = 0$, $\lambda = 1$

$x = \frac{1}{3}$

Substitute $y = 0$ in eq^n ①.

$2x^2 + 0 = 1$

$x^2 = \frac{1}{2}$

$x = \pm \frac{1}{\sqrt{2}}$, $\left(\frac{1}{\sqrt{2}}, 0\right)\left(-\frac{1}{\sqrt{2}}, 0\right)$

Substitute $x = 1/3$ in eq^n ①.

$2\left(\frac{1}{3}\right)^2 + y^2 = 1$

$\frac{2}{9} + y^2 = 1$

$y^2 = 1 - \frac{2}{9} = \frac{9-2}{9} = \frac{7}{9}$

$y = \pm\frac{\sqrt{7}}{3}$, $\left(\frac{1}{3}, \frac{\sqrt{7}}{3}\right)\left(\frac{1}{3}, -\frac{\sqrt{7}}{3}\right)$

(x,y)	$\left(\frac{1}{\sqrt{2}}, 0\right)$	$\left(-\frac{1}{\sqrt{2}}, 0\right)$	$\left(\frac{1}{3}, \frac{\sqrt{7}}{3}\right)$	$\left(\frac{1}{3}, -\frac{\sqrt{7}}{3}\right)$
$f(x,y) = 4x^3 + y^2$	$4\left(\frac{1}{\sqrt{2}}\right)^3$	$4\left(-\frac{1}{\sqrt{2}}\right)^3$	$4\left(\frac{1}{3}\right)^3 + \left(\frac{\sqrt{7}}{3}\right)^2$	$4\left(\frac{1}{3}\right)^3 + \left(-\frac{\sqrt{7}}{3}\right)^2$
	$\sqrt{2}$	$-\sqrt{2}$	$25/27$	$25/27$
	↓ maximum	↓ minimum	other stationary values.	

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8. $f(x,y) = x - 3y - 1 \; ; \; x^2 + 3y^2 = 16.$

Solve

$f(x,y) = x - 3y - 1 \longrightarrow ①$

$g(x,y) = x^2 + 3y^2 - 16 \longrightarrow ②$

$\nabla f = \lambda \nabla g$

$f_x = \lambda g_x \quad , \quad f_y = \lambda g_y$

$1 = \lambda 2x \quad , \quad -3 = \lambda 6y$

$\lambda = \frac{1}{2x} \quad , \quad \lambda = \frac{-1}{2y}$

$\lambda = \lambda$

$\frac{1}{2x} = \frac{-1}{2y}$

$y = -x$

Put $y = -x$ in ②.

$x^2 + 3(-x)^2 - 16$

$4x^2 = 16$

$x = \pm 2$

Points : $(2, -2)$, $(-2, 2)$

(x,y)	$(2,-2)$	$(-2,2)$
$f(x,y) =$	$2 - 3(-2) - 1$	$-2 - 3(2) - 1$
$x - 3y - 1$	7	-9
	↓	↓
	maximum	minimum

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9. $f(x,y,z) = 2x+y-2z \; ; \; x^2+y^2+z^2 = 4$

Solve

$f = 2x+y-2z \longrightarrow ①$

$g = x^2+y^2+z^2-4 \longrightarrow ②$

$\nabla f = \lambda \nabla g$

$f_x = \lambda g_x \quad , \quad f_y = \lambda g_y \quad , \quad f_z = \lambda g_z$

$2 = \lambda 2x \quad , \quad 1 = \lambda 2y \quad , \quad -2 = \lambda 2z$

$\lambda = \frac{1}{x} \quad , \quad \lambda = \frac{1}{2y} \quad , \quad \lambda = -\frac{1}{z}$

$\frac{1}{x} = \frac{1}{2y} \quad , \quad \frac{1}{x} = -\frac{1}{z}$

$x = 2y \quad , \quad x = -z$

Put the values in eqn ②.

$x^2 + \frac{x^2}{4} + x^2 = 4$

$9x^2 = 16$

$x^2 = 16/9$

$\boxed{x = \pm \frac{4}{3}}$

$y = \frac{x}{2}$

$\boxed{y = \pm \frac{2}{3}}$

$z = -x$

$\boxed{z = \pm \frac{4}{3}}$

Points : $\left(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3}\right), \left(-\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right)$

(x,y,z)	$\left(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3}\right)$	$\left(-\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right)$
$f(x,y,z)$	$18/3$	$-18/3$
	6	-6
	↓ maximum	↓ minimum

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10. $f(x,y,z) = 3x+6y+2z$; $2x^2+4y^2+z^2 = 70$

Solve

$f(x,y,z) = 3x+6y+2z \longrightarrow ①$

$g(x,y,z) = 2x^2+4y^2+z^2 = 70 \longrightarrow ②$

$\nabla f = \lambda \nabla g$

$f_x = \lambda g_x \quad , \quad f_y = \lambda g_y \quad , \quad f_z = \lambda g_z$

$3 = \lambda 4x \quad , \quad 6 = \lambda 8y \quad , \quad 2 = \lambda 2z$

$x = \frac{3}{4\lambda} \quad , \quad y = \frac{3}{4\lambda} \quad , \quad z = \frac{1}{\lambda}$

Put the values in eqⁿ ②.

$$2\left(\frac{3}{4\lambda}\right)^2 + 4\left(\frac{3}{4\lambda}\right)^2 + \left(\frac{1}{\lambda}\right)^2 = 70$$

$$\frac{18}{16\lambda^2} + \frac{36}{16\lambda^2} + \frac{1}{\lambda^2} = 70$$

$$\frac{18+36+16}{16\lambda^2} = 70 \quad \Rightarrow \frac{70}{16\lambda^2} = 70$$

$$16\lambda^2 = 1$$

$$\lambda = \pm \frac{1}{4}$$

Case 1: $\lambda = 1/4$

$$x = \frac{3}{4(1/4)} = 3 \quad , \quad y = \frac{3}{4(1/4)} = 3 \quad , \quad z = \frac{1}{(1/4)} = 4$$

Point : $(3, 3, 4)$.

Case 2: $\lambda = -1/4$

$x = -3 \quad , y = -3 \quad , z = -4$

Point : $(-3, -3, -4)$

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(x,y,z)	$(3,3,4)$	$(-3,-3,-4)$
$f(x,y,z)$	$3(3)+6(3)+2(4)$	$3(-3)+6(-3)+2(-4)$
	$9+18+8$	$-9-18-8$
	35	-35
	↓	↓
	maximum	minimum

11. $f(x,y,z) = xyz \; ; \; x^2+y^2+z^2 = 1$

solve

$f(x,y,z) = xyz \longrightarrow ①$

$g(x,y,z) = x^2+y^2+z^2-1 \longrightarrow ②.$

$\nabla f = \lambda \nabla g$

$f_x = \lambda g_x \quad , \quad f_y = \lambda g_y \quad , \quad f_z = \lambda g_z$

$yz = \lambda 2x \quad , \quad xz = \lambda 2y \quad , \quad xy = \lambda 2z$

x by x , x by y , x by z

$xyz = \lambda 2x^2 \quad , \quad xyz = \lambda 2y^2 \quad , \quad xyz = \lambda 2z^2$

$\lambda 2x^2 = \lambda 2y^2 = \lambda 2z^2$

If $\lambda \neq 0$, then

$x^2 = y^2 = z^2$

Substitute this into eq^n ②.

$x^2 + x^2 + x^2 = 1$

$3x^2 = 1$

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$$x^2 = \frac{1}{3}$$

$$x = \pm \frac{1}{\sqrt{3}}$$

This give 8 points: $\left(\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}\right)$

Plug points into eqn ①.

If a point has odd no of - signs (one or three):

$$f\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) = -\frac{1}{3\sqrt{3}} \longrightarrow \text{minimum}$$

If a point has even no of - signs (0 or two):

$$f\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) = \frac{1}{3\sqrt{3}} \longrightarrow \text{maximum}$$

Ch#14 - Multiple Integrals

Ex#14.1 - Double Integrals

Double Integral = Volume under the surface

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